Assignment 5

Database Creation and Table Implementation

MySQL DDL commands and finalized relationship structure

Project: LibSphere - Library Management System **Prepared:** 23 October 2026

Platform assumption: MySQL 8.0.16 or later (InnoDB). Outputs in this report are expected results derived from the included sample dataset; run the SQL in MySQL Workbench to capture authentic execution screenshots if required by the instructor.

This implementation creates the LibSphere database with six core tables. Run the statements in the order shown because child tables use foreign keys to reference parent tables. The full script is also provided as a separate .sql file in the submission bundle.

1. Relationship overview

- Categories (1) to Books (many): every book references one category.
- Books (many) to Authors (many): resolved with Book_Authors, whose composite primary key is (book_id, author_id).
- Members (1) to Loans (many): each loan belongs to one member.
- Books (1) to Loans (many): each loan references one book title; a title can appear in many historical transactions.

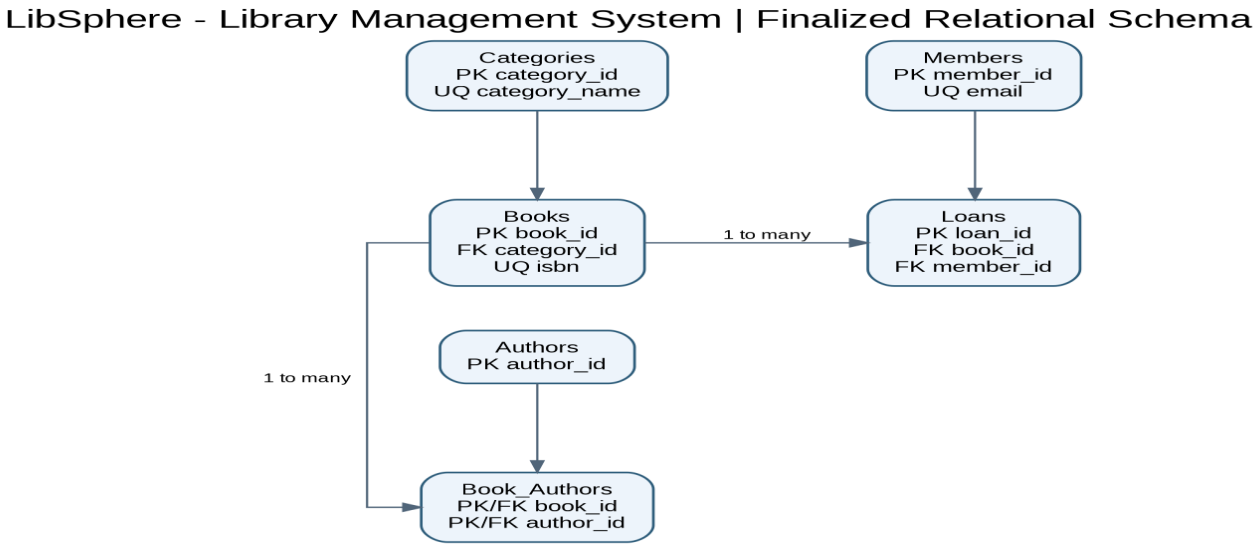


Figure 1. Logical relationship layout used by the DDL below.

2. Create the database and parent tables

```
CREATE DATABASE IF NOT EXISTS libsphere;
USE libsphere;

CREATE TABLE Categories (
    category_id INT AUTO_INCREMENT PRIMARY KEY,
    category_name VARCHAR(50) NOT NULL UNIQUE
) ENGINE=InnoDB;

CREATE TABLE Authors (
    author_id INT AUTO_INCREMENT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    bio TEXT NULL
) ENGINE=InnoDB;
```

3. Create Books and Book_Authors

```
CREATE TABLE Books (
    book_id INT AUTO_INCREMENT PRIMARY KEY,
    title VARCHAR(150) NOT NULL,
    isbn VARCHAR(13) NOT NULL UNIQUE,
    category_id INT NOT NULL,
    publication_year INT NOT NULL,
    total_copies INT NOT NULL DEFAULT 1,
    available_copies INT NOT NULL DEFAULT 1,
    CONSTRAINT chk_books_isbn_length CHECK (CHAR_LENGTH(isbn) = 13),
    CONSTRAINT chk_books_year CHECK (publication_year BETWEEN 1450 AND 2100),
    CONSTRAINT chk_books_total_copies CHECK (total_copies >= 1),
    CONSTRAINT chk_books_available_copies CHECK
        (available_copies >= 0 AND available_copies <= total_copies),
    CONSTRAINT fk_books_category FOREIGN KEY (category_id)
        REFERENCES Categories(category_id)
        ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB;
```

```
CREATE TABLE Book_Authors (
    book_id INT NOT NULL,
    author_id INT NOT NULL,
    PRIMARY KEY (book_id, author_id),
    CONSTRAINT fk_book_authors_book FOREIGN KEY (book_id)
        REFERENCES Books(book_id) ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_book_authors_author FOREIGN KEY (author_id)
        REFERENCES Authors(author_id) ON UPDATE CASCADE ON DELETE CASCADE
) ENGINE=InnoDB;
```

4. Create Members and Loans

```
CREATE TABLE Members (
    member_id INT AUTO_INCREMENT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE,
    phone VARCHAR(20) NOT NULL,
    membership_date DATE NOT NULL,
    status VARCHAR(20) NOT NULL DEFAULT 'Active',
    CONSTRAINT chk_members_status CHECK
        (status IN ('Active', 'Suspended', 'Expired'))
) ENGINE=InnoDB;

CREATE TABLE Loans (
    loan_id INT AUTO_INCREMENT PRIMARY KEY,
    book_id INT NOT NULL,
    member_id INT NOT NULL,
    loan_date DATE NOT NULL,
```

```
    due_date DATE NOT NULL,
    return_date DATE NULL,
    status VARCHAR(20) NOT NULL DEFAULT 'Issued',
    CONSTRAINT chk_loans_status CHECK
        (status IN ('Issued', 'Returned', 'Overdue')),
    CONSTRAINT chk_loans_due_date CHECK (due_date >= loan_date),
    CONSTRAINT chk_loans_return_date CHECK
        (return_date IS NULL OR return_date >= loan_date),
    CONSTRAINT chk_loans_status_return CHECK
        ((status = 'Returned' AND return_date IS NOT NULL) OR
         (status IN ('Issued', 'Overdue') AND return_date IS NULL)),
    CONSTRAINT fk_loans_book FOREIGN KEY (book_id)
        REFERENCES Books(book_id) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_loans_member FOREIGN KEY (member_id)
        REFERENCES Members(member_id) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB;
```

5. Design choices

Choice	Purpose
InnoDB	Enables transactional behavior and foreign-key enforcement.
AUTO_INCREMENT integer primary keys	Provides stable identifiers without depending on names or other editable attributes.
Book_Authors composite key	Stops a repeated book-author link while supporting co-authored books.
RESTRICT on Loans parent references	Protects circulation history from being removed by deleting a member or book with a loan record.
CASCADE on bridge references	Allows obsolete Book_Authors link rows to be removed when a parent is deleted; Loans remain protected.
CHECK constraints	Reject invalid inventory counts, implausible year values, and invalid status/date combinations.

Version note: CHECK constraints are enforced in MySQL 8.0.16 and later. Use the same six-table schema in Assignments 6–9 so all outputs and relationships match.